

Efficient Drone Exploration in Real Unknown Environments

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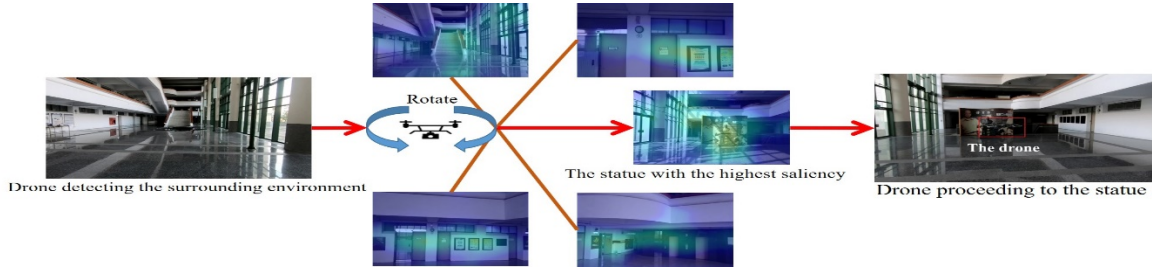


Figure 1: Overview of the proposed system: drone detecting the surrounding environment and proceeding to the statue.

ABSTRACT

We propose an autonomous drone exploration system (ADES) with a lightweight and low-latency saliency prediction model to explore unknown environments. Recent studies have applied saliency prediction to drone exploration. However, these studies are not sufficiently mature. The ADES system proposes a smaller and faster saliency prediction model and adopts a novel drone exploration approach based on visual-inertial odometry (VIO) to solve the practical problems encountered during exploration, i.e., exploring salient objects without colliding with them and not repeatedly exploring salient objects. The system not only has a performance comparable to that of the state-of-the-art multiple-discontinuous-image saliency prediction network (TA-MSNet) but also enables drones to explore unknown environments more efficiently.

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1 INTRODUCTION

Recently, the use of drones for exploration has become emerging research; however, these studies have many drawbacks. Dang et al. [Dang et al., 2018] proposed an application for drones exploring the environment based on saliency. The application requires considerable time to build a map to plan the exploration path, and they used a saliency prediction model that lacks time and the whole environment information. Chu et al. [Chu et al., 2021] proposed the first

saliency prediction network (MS-Net) considering the whole environment information. Although MS-Net can predict the saliency after taking multiple discontinuous images of the whole environment, the predictions of MS-Net do not consider temporal information. This may lead to inconsistent prediction results for different time frames. Huang et al. [Huang et al., 2022] improved MS-Net by proposing TA-MSNet, which considers both the whole environment information and the time information. However, TA-MSNet is slower and larger, which is less suitable for use in real drone exploration. Moreover, MS-Net and TA-MSNet only conducted simulations on image datasets, and they did not actually apply the network to drones for exploration field tests to prove the feasibility of their networks.

In this paper, we propose the Faster-TA-MSNet and integrate a novel drone exploration approach based on visual-inertial odometry (VIO) to construct an autonomous drone exploration system (ADES). The proposed Faster-TA-MSNet improves the TA-MSNet, making it faster, lighter, and more suitable for real-time computing on drones, and reaches similar performance to that of TA-MSNet. The proposed drone exploration approach fuses saliency, IMU data, compass data, and visual SLAM to control the drone in unknown environments. Therefore, the proposed ADES enables the drone to, just like human beings, first observe the surrounding environment in an unknown environment and then predict the saliency based on multiple discontinuous images to determine the exploration direction (Figure 1). Then, the drone explores salient objects, avoids the colliding salient objects, and records the explored locations. This cycle continues until the entire environment has been explored.

2 SYSTEM ARCHITECTURE

Figure 2 shows the system architecture of the proposed ADES. ADES was a modification of the drone control system proposed by [Chen et al., 2022]. It contains two parts: Faster-TA-MSNet and the

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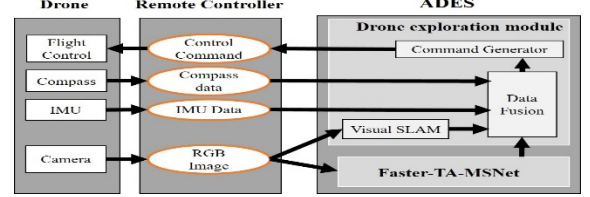
Table 1: Comparison results between proposed Faster-TA-MSNet and TA-MSNet.

	Proposed	TA-MSNet
CPU Inference Time (s)	6.0	6.4
Number of Parameters (million)	57.48	108.07
CC ↑	0.1997	0.2075
SIM ↑	0.2483	0.2538
KL ↓	1.2218	1.2296

drone exploration module. The Faster-TA-MSNet improved the TA-MSNet. First, we replaced the backbone EML-Net with EfficientNet-b0 [Tan et al., 2019] to accelerate the computational speed of the network. In addition, we replaced the TA-MSNet transformer with a graph convolutional network [Zhang et al., 2020]. Faster-TA-MSNet analyzed the images taken by the drone in five different directions and then predicted and output their saliency values to the drone exploration module.

The drone exploration module included visual SLAM, data fusion, and command generator. The visual SLAM used by ADES was ORB-SLAM2 [Mur-Artal et al., 2017], a system that performed real-time SLAM based on the feature points in the image. It could generate a sparse 3D scene based on the input image and calculate the camera trajectory. ADES inputs real-time drone images to ORB-SLAM2 to calculate and record the drone trajectory. The data fusion involved fusing the saliency values from Faster-TA-MSNet, IMU data, compass data, and drone trajectory to the command generator. Finally, the flight control of the drone received the control commands generated by the command generator and used them to control the drone.

The workflow of ADES could be described as follows: ADES first rotated the drone and observed its surroundings, and five images with different orientations and the corresponding drone heading angles were acquired during the rotation. These five images, live images of the drone, compass data (drone heading angles), and IMU data (drone’s velocity) were then input into ADES. After receiving the analysis results from Faster-TA-MSNet, the drone exploration module controlled the drone to turn toward and fly toward the object with the highest saliency value based on the object’s saliency value and the corresponding drone heading angle. The images of the flight process were sent to ORB-SLAM2, and the drone exploration module first compared the current flight trajectory output from ORB-SLAM2 with the previously recorded trajectories. If the similarity among the current trajectory and the previous trajectories was high, this salient object had been explored. The system then stopped the drone and controlled the drone to turn and head to the next higher salient object. The drone stopped when it reached a salient object because the current commercial drones were equipped with an obstacle avoidance system, and we used the obstacle avoidance function to stop the drone in front of a salient object. At this point, when ADES had the drone’s velocity approaching 0, the system rotated the drone again and took five images of the surrounding environment to start the new round of significant objects to be explored. This cycle continued until the drone explored the entire environment. A demo video of ADES can be found on the following link: <https://youtu.be/uuTzpWdUf3U>.

**Figure 2: ADES system architecture.**

3 EXPERIMENT

In order to evaluate the proposed model, we compared it with the state-of-the-art method (TA-MSNet) from three aspects. Table 1 shows the comparison results. With respect to the CPU inference time, the proposed model was 6% faster than TA-MSNet. In terms of the number of parameters, that of the proposed model was 47% smaller than that of TA-MSNet. For comparing performance, we used the same dataset in TA-MSNet for training and testing and used the same metrics: similarity (SIM), correlation coefficient (CC), and Kullback–Leibler divergence (KL). The results revealed that the proposed model and TA-MSNet had very similar performance.

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